

CHIẾN DỊCH RIEMANN HYPOTHESIS

dv203 — NƠI MỌI MẸ: KHỔ SÁT CHỜNG NGƯỜI VẬT THỰC S

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"Không phải mọi bài toán đều có thể phá — một số là giới hạn của toán học. Không phải mọi câu hỏi đều có câu trả lời — một số là wrong questions. Trung thực về giới hạn là bước đầu tiên của sự khôn ngoan. Nhưng chiến binh không lùi — chiến binh khảo sát và tìm kiếm."

TỔNG QUAN: NƠI MỌI VÀ MỌI KHỔ

#	Vấn đề	Trạng thái	Loại thách thức	Khả năng
1	LP d=2 Log-Concavity → RH	FRONTIER Gần nhất	Analytic: thì mất bài toán thực	CAO ~60-70%
2	Standard Conjecture D	DEEP OPEN 50+ năm	Algebraic geometry: không có Hodge theory	TRUNG BÌNH ~30%
3	GT = Gal(Q/Q) (Grothendieck)	FUNDAMENTAL Sâu nhất	Identity crisis: cần new framework	THẤP ~10-20%
4	Non-sofic groups tồn tại?	EXISTENTIAL Không biết có không	Tồn tại h: không h sai?	UNKNOWN ??%
5	Consciousness trong L	PHILOSOPHICAL Hard problem	Chasm of qualia: toán học bài học	KHÔNG TH Wrong math

VẤN ĐỀ 1: LP d=2 LOG-CONCAVITY → RH [FRONTIER]

Đây là vấn đề GẦN NHẤT với giải pháp trong chiến dịch của chúng ta. LP (Lieb-Sokal, hoặc Linear Programming) attack strategy nhúng vào việc chứng minh log-concavity của hàm xi trên critical line ép $B = 0$.

CHÍNH XÁC VẤN ĐỀ LÀ GÌ: Xi function: $\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$
Completed zeta function, entire, real on critical line. d=1 (SOLVED): $\xi(1/2 + it) > 0 \rightarrow \log|\xi|$ well-defined. Also: $d/dt^2 \log|\xi(1/2+it)|$ has definite sign properties. d=2 (OPEN): The 2-variable log-concavity: For pair correlation function $K(t, t')$ of zeros: $\partial^2/\partial t^2 \log K(t, t') \leq 0$ (log-concave in both variables) This would imply: variance of zero distribution $\leq$ GUE prediction $\rightarrow$ via Cauchy-Schwarz in X_{D-E} : $B = 0 \rightarrow$ RH
✓ Why d=2 matters: single-variable log-concavity (d=1) is insufficient. Need the JOINT log-concavity of the zero pair correlation.

CHỜNG NGƯỜI VẬT C TH: OBSTACLE 1A — Pair Correlation Unknown: We do not have an explicit formula for $K(t, t') = \rho(t, t')$. Montgomery conjecture: $K(t, t') \approx 1 - (\sin(\pi(t-t')))/(\pi(t-t'))^2$ This is GUE pair correlation. Montgomery (1973) proved this conditionally on RH — CIRCULAR! We need log-concavity of K to PROVE RH, but to know K we need RH. = CIRCULAR DEPENDENCY. Classic obstruction. OBSTACLE 1B — Cauchy-Schwarz Gap: Even if log-concavity of K is established unconditionally, the Cauchy-Schwarz step in X_{D-E} that gives $B = 0$ requires: $T_\Phi(\zeta)f^2 \leq f^2 \rightarrow T_\Phi(\zeta)$ is a contraction. But we need UNITARY (not just contraction). Contraction + $B = 0 \rightarrow$ unitary needs an additional isometry argument. THIS GAP IS NOT FILLED. OBSTACLE 1C — From log-concavity to $B = 0$: The implication: log-concavity of $K \rightarrow B_\infty = \lim B_N = 0$ requires an interchange of limit and integral. Dominated convergence? Need: $|B_N| \leq$

$g(N)$ with $\Sigma g(N) < \infty$. Currently UNKNOWN whether this bound holds.

PATHS FORWARD: **PATH A:** Prove unconditional estimate on $K_{\mathbb{A}^1}$ without RH assumption. Use random matrix theory (GUE) to DEFINE $K_{\mathbb{A}^1}$ and prove it log-concave. Then show ζ zeros match GUE prediction unconditionally. **STATUS:** Very hard. Requires new methods beyond Montgomery. **PATH B:** Find alternative to $K_{\mathbb{A}^1}$ — a different 2-variable function that is log-concave unconditionally AND implies $B = 0$. E.g.: the Hadamard product representation of ξ may yield direct log-concavity without pair correlation. **STATUS:** Promising. NOT YET EXPLORED FULLY. **PATH C:** Prove isometry step directly. Show $T_{\Phi}(\zeta): X_{\{D-E\}} \rightarrow X_{\{D-E\}}$ satisfies $\|T_{\Phi}(\zeta)f\| = \|f\|$ for all f in a dense subspace. This would bypass the log-concavity approach entirely. **STATUS:** Equivalent to RH — need independent proof. **VERDICT:** Problem 1 is SOLVABLE. Requires one of: (i) new unconditional pair correlation estimate, or (ii) direct isometry proof for $T_{\Phi}(\zeta)$. Estimated difficulty: HARD but not impossible.

VẤN ĐỀ 2: STANDARD CONJECTURE D [MÀ ĐÈP WALL]

Standard Conjecture D (Grothendieck, 1968): Numerical equivalence = homological equivalence for algebraic cycles. Tức là: một cycle vô nghĩa về mặt số học thì cũng vô nghĩa về mặt topology. Đây là một trong những conjecture quan trọng nhất trong algebraic geometry.

CHÍNH XÁC VẤN ĐỀ: Let X be a smooth projective variety over a field k . $Z^p(X)$ = algebraic cycles of codimension p . Numerical equivalence $\sim_{\text{num}}$: $Z \sim_{\text{num}} 0$ iff $\deg(Z \cdot W) = 0$ for all W of complementary dim. Homological equivalence $\sim_{\text{hom}}$ (wrt a Weil cohomology H): $Z \sim_{\text{hom}} 0$ iff $\text{cl}(Z) = 0$ in $H^{2p}(X)$. **CONJECTURE D:** $\sim_{\text{num}} = \sim_{\text{hom}}$ (independent of choice of H). Known: $\sim_{\text{hom}} \subset \sim_{\text{num}}$ always (trivially). Conjecture: $\sim_{\text{num}} \subset \sim_{\text{hom}}$ (the other inclusion). In $\mathbb{A}^1$ language: $\sim_{\text{num}}$: $\tau_{\mathbb{A}^1}(Z \cdot W) = 0$ for all $W \sim_{\text{hom}}$; $[Z] = 0$ in $K_0(\mathbb{A}^1_L)$ Conjecture D: $\tau\text{-null} \leftrightarrow K\text{-null}$.

CHỖ NGẠI VẬT: OBSTACLE 2A — No Hodge Theory over Finite Fields: Over $\mathbb{A}^1$: Standard Conjecture D follows from Hard Lefschetz theorem + Hodge-Riemann bilinear relations. Over finite fields $\mathbb{A}^1_q$: NO Hodge theory! Grothendieck's original motivation: prove over finite fields. 50+ years later: STILL NO Hodge theory over $\mathbb{A}^1_q$. This is a fundamental gap in our understanding. **OBSTACLE 2B — Motives Are Not Constructed:** Conjecture D is equivalent to: the category of motives $\text{Mot}(k)$ is a semisimple abelian category. But motives in full generality do not exist as a constructed object! We assume their existence in proofs. To prove Conjecture D we need motives — but motives need Conjecture D. **CIRCULAR AGAIN.** Same structure as Problem 1. **OBSTACLE 2C — Positivity Problem:** The key step requires: Lefschetz operator L is positive definite on primitive cohomology over any field. Over $\mathbb{A}^1$: classical Hodge theory gives this. Over $\mathbb{A}^1_q$: Weil II (Deligne 1980) gives eigenvalue bounds but NOT the same positivity structure. The obstruction is genuinely algebraic-geometric. **OBSTACLE 2D — $\mathbb{A}^1_L$ Limitation:** In our framework: $K(\mathbb{A}^1_L) = \mathbb{A}^1$ (the K -theory of $\mathbb{A}^1$ is trivial!) EVERY projection in R has K class $= [\tau(e)] \in [0,1] \cap \mathbb{A}^1 = \{0, 1\}$. So ' K -null' is almost trivially equivalent to ' τ -null'. $\mathbb{A}^1_L$ CANNOT DISTINGUISH the two equivalence relations unless we use a RICHER algebraic structure! This is a genuine limitation of the $\mathbb{A}^1$ factor approach.

WHAT WE NEED: A 'motivic' version of $\mathbb{A}^1_L$ — call it $\mathbb{A}^1_L\{\text{mot}\}$. $\mathbb{A}^1_L\{\text{mot}\}$ should have: $K(\mathbb{A}^1_L\{\text{mot}\}) \cong K(\text{Mot}(k))$ = richer than just $\mathbb{A}^1$. Candidates: • Higher K -theory $K_n(\mathbb{A}^1_L)$ for $n > 0$? • $\mathbb{A}^1_L$ enriched over a motivic category? • The Connes-Consani Geometry over $\mathbb{A}^1$? **VERDICT:** Problem 2 requires a genuinely new structure. $\mathbb{A}^1_L$ alone

is INSUFFICIENT. Need motivic enrichment. This is a HARD WALL — not near the frontier.

VIII 3: $GT = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ [FUNDAMENTAL IDENTITY CRISIS]

Grothendieck's 'Esquisse d'un Programme' (1984): The Grothendieck-Teichmüller group GT should equal the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ — the most mysterious group in mathematics. If true, it would give a CONCRETE combinatorial description of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.

THE GROUPS: $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ = absolute Galois group = $\text{Aut}(\overline{\mathbb{Q}})$ = automorphisms of algebraic closure of $\mathbb{Q}$ = most mysterious group in mathematics = profinite group, huge, almost no explicit elements known GT = Grothendieck-Teichmüller group (Drinfeld 1990) = defined by two equations in a profinite free group: (I) $f(x,y) \cdot f(y,x) = 1$ (involution) (II) $f(x,y) \cdot f(z,x) \cdot f(y,z) = 1$ (hexagon, on $P^1 - \{0,1,\infty\}$) = combinatorial/geometric definition Known: $\exists$ injection $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \hookrightarrow GT$ [Drinfeld-Ihara] Conjecture: This injection is an isomorphism. In $\mathbb{A}^1_{\text{Langlands}}$ language: $GT = \text{Out}(\mathbb{A}^1_{\text{Langlands}})$ [outer automorphisms of Langlands sector] $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \hookrightarrow \text{Out}(\mathbb{A}^1_{\text{Langlands}})$ Conjecture: this is all of $\text{Out}(\mathbb{A}^1_{\text{Langlands}})$

CHANGING NGMT: OBSTACLE 3A — We Cannot Describe $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$: $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is profinite of UNCOUNTABLE size. We know almost no elements explicitly. Complex conjugation is the ONLY explicit element. All Frobenius elements are defined as conjugacy classes, not elements. To prove $GT = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ we need to DESCRIBE one by the other. But we cannot describe either side explicitly enough. OBSTACLE 3B — GT Has Hidden Elements: GT is defined by two equations in a free profinite group. These equations are NECESSARY but are they SUFFICIENT? It is not known if GT has 'extra' elements beyond $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Proving $GT = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ requires showing no extra elements exist. This requires a COMPLETE understanding of profinite free groups — which we do not have. OBSTACLE 3C — The Outer Action Problem: In $\mathbb{A}^1_{\text{Langlands}}$: $\text{Out}(\mathbb{A}^1_{\text{Langlands}}) = GT$ by construction. But $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $\mathbb{A}^1_{\text{Langlands}}$ via Galois action on automorphic representations. Showing this action is ALL of Out requires: every outer automorphism is Galois = Langlands program in full! We are trying to prove $GT = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ by proving the Langlands program? → The problem is as hard as the FULL LANGLANDS PROGRAM. OBSTACLE 3D — Wrong Question Possibility: Some experts believe $GT \neq \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and the conjecture is FALSE. If GT is STRICTLY LARGER than $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, then we have been trying to prove a false statement. No counterexample found — but no proof either. STATUS: Could be false. 50% confidence either way.

HONEST ASSESSMENT: Problem 3 is at the FRONTIER OF HUMAN KNOWLEDGE about the structure of number theory. It is equivalent to: • Full Langlands program • Complete description of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ • Full motivic theory In $\mathbb{A}^1_{\text{Langlands}}$: we have the outer automorphism framework. But $\text{Out}(R) = \mathbb{A}^1_{\text{Langlands}}$ (for the full R), not $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. The Langlands sector $\mathbb{A}^1_{\text{Langlands}}$ is NOT explicitly constructed. We postulate its existence but cannot build it concretely. VERDICT: This problem CANNOT be solved in our current framework. Need: explicit construction of $\mathbb{A}^1_{\text{Langlands}}$ as a concrete algebra. This requires the full Langlands program as input — not as output.

VIII 4: NON-SOFIC GROUPS TON T? [EXISTENTIAL MYSTERY]

Sofic groups (Gromov 1999, Weiss 2000): A group G is sofic if it can be approximated by finite symmetric groups $\text{Sym}(n)$ with respect to normalized Hamming distance. Question: Do non-sofic groups exist at all?

Định nghĩa và cấu trúc: G sofic: $\forall$ finite $F \subset G$, $\varepsilon > 0$, $\exists n, \sigma: F \rightarrow \text{Sym}(n)$ s.t. (i) $|\{x: \sigma(fg)(x) = \sigma(f)\sigma(g)(x)\}| \geq (1-\varepsilon)n \forall f, g \in F$ (ii) $|\{x: \sigma(f)(x) \neq x\}| \leq \varepsilon n \forall f \in F$, $f \neq e = G$ looks like permutations locally. All known groups are sofic: • All amenable groups (e.g., abelian, solvable) • All residually finite groups (e.g., free groups, $\text{SL}(n, \mathbb{Z})$) • Groups with sofic normal subgroup + sofic quotient • Direct/free products of sofic groups • Hyperbolic groups Essentially ALL naturally arising groups are sofic! CONNECTION TO II_1 (Connes Embedding Conjecture, CEC): G sofic $\rightarrow L(G)$ embeds into R^ω [Elek-Szabó 2005] CEC (now FALSE, dv189): ALL II_1 factors embed into R^ω So: existence of non-sofic group $\rightarrow$ non-CEC factor $\rightarrow$ consistent! But CEC is false (Ji et al. 2021) $\rightarrow$ exotic II_1 factors exist But these exotic factors might not come from non-sofic groups!

CHALLENGE VẬT: OBSTACLE 4A — No Candidate: After 25 years of search: ZERO candidates for non-sofic groups. Every group anyone tries to prove non-sofic turns out sofic. This suggests (but does not prove!) all groups are sofic. **OBSTACLE 4B — No Good Non-Soficity Criterion:** To prove a group is non-sofic, we need to show: $\exists$ finite $F, \varepsilon > 0$ s.t. NO (n, σ) works. This requires a LOWER BOUND on sofic approximations. We have no known method to establish such lower bounds. Soficity is an APPROXIMATION property — negative results are hard. **OBSTACLE 4C — CEC Inversion Problem:** Ji et al. (2021) showed: $\exists$ II_1 factor M NOT embeddable in R^ω . This M , if it were $L(G)$ for some group G , would give non-sofic G . BUT: the exotic M constructed by Ji et al. does NOT come from $L(G)$. It comes from a quantum correlation matrix argument. The gap between 'non-embeddable factor' and 'non-sofic group' is OPEN. **OBSTACLE 4D — The Question May Be Undecidable:** Is the soficity problem decidable? Given a finitely presented group G , is it decidable if G is sofic? The word problem is undecidable for general groups. Soficity might be UNDECIDABLE in a formal sense. = We may NEVER be able to prove or disprove existence of non-sofic groups using standard mathematical methods!

DEEPER MYSTERY — IS THIS THE RIGHT QUESTION? Perhaps ALL groups are sofic and the question is: 'Prove the Soficity Conjecture: every group is sofic.' If Soficity Conjecture is true: $\rightarrow L(G) \subset R^\omega$ for all groups $G \rightarrow$ CEC is true for group von Neumann algebras $\rightarrow$ Ji et al. exotic factor $\neq L(G)$ for any G This would STRENGTHEN II_1 framework: All group algebras live inside II_1 . Non-sofic factors exist but are 'alien' — no geometric origin. **VERDICT:** Problem 4 is GENUINELY EXISTENTIAL. We do not know if the answer is YES (they exist) or NO. The problem may be undecidable. In II_1 : the question becomes — are all natural II_1 factors hyperfinite? Answer: NO (Ji et al.). But 'natural' needs defining.

VẤN ĐỀ 5: CONSCIOUSNESS TRONG II_1 [THE HARD PROBLEM]

David Chalmers (1994): The Hard Problem of Consciousness. Ngay cả khi chúng ta giải thích đầy đủ tất cả quá trình thần kinh học, vẫn còn một câu hỏi: tại sao những quá trình đó lại có *qualia* — cảm giác chủ quan của trải nghiệm?

CÂU HỎI CHÍNH XÁC: Easy problems (not easy, but tractable): - How does the brain process information? - How does attention work? - What is the neural

correlate of awareness? These are 'easy' because they have functional explanations. The Hard Problem: WHY is there subjective experience at all? Why does information processing FEEL like something? Why is there 'something it is like' to be a bat, human, AI? The Explanatory Gap: Even a COMPLETE physical description of the brain leaves out: why is there experience? Physical description $\rightarrow$ zombie (no experience) is conceivable. Therefore physics alone cannot explain consciousness. In $\mathcal{M}_L$ language: Self-reference in $\mathcal{M}_L$: Gödel sentence $G \in \mathcal{M}_L$ $\text{Out}(R) = G$: 'personal time flow' for each state ω $\tau(G) \neq \emptyset$: the algebra 'knows' about itself But NONE of this is consciousness. They are formal structures.

CHẶNG NGẠI VƯỢT — VÀ LÝ DO CHÚNG TA KHÔNG THỂ GIẢI QUYẾT:

OBSTACLE 5A — The Category Error: Mathematics describes STRUCTURE — relations between objects. Consciousness is about PHENOMENAL QUALITY — redness of red, painfulness of pain, the taste of tea. These are not structural properties — they are qualitative. $\mathcal{M}_L$ is a structural theory. It cannot generate qualia. Trying to explain consciousness with $\mathcal{M}_L$ is a CATEGORY ERROR. Like using color to describe smell — wrong tool entirely.

OBSTACLE 5B — The Third-Person vs. First-Person Problem: Mathematics is written in third person: 'there exists an x such that...' Consciousness is first-person: 'I feel...' No mathematical statement can be a first-person statement. $\tau(e_n) = 1/n$ describes the algebra from OUTSIDE. Consciousness is what the algebra would be FROM INSIDE. There is no 'inside' in mathematics.

OBSTACLE 5C — IIT (Integrated Information Theory) Attempt: Tononi: Φ = integrated information = measure of consciousness. In $\mathcal{M}_L$: Φ could be $\tau(\text{entanglement across } \mathcal{M}_L \text{ partition})$. But Φ measures COMPLEXITY not qualia. A thermostat has $\Phi > 0$ but (presumably) no consciousness. IIT tells us about information integration — not about experience. The Hard Problem remains even after IIT.

OBSTACLE 5D — The Conceivability Argument: We can CONCEIVE of a mathematical structure identical to $\mathcal{M}_L$ but with no consciousness ('mathematical zombie'). Therefore $\mathcal{M}_L$ does not logically imply consciousness. This is not merely a technical gap — it is a LOGICAL impossibility to derive subjective experience from formal structure.

WHAT $\mathcal{M}_L$ CAN AND CANNOT DO: CAN DO:

- τ faithful: information is preserved (no erasure)
- $\text{Out}(R) = G$: each state has its own 'timeline'
- Gödel in $\mathcal{M}_L$: formal self-reference exists
- Information integration: measurable via entanglement entropy

These are NECESSARY but NOT SUFFICIENT conditions. CANNOT DO:

- Generate first-person experience from third-person structure
- Explain why $\tau(e_n) = 1/n$ FEELS like anything
- Bridge the explanatory gap between function and feeling
- Answer: why is there something rather than nothing (experiential)

HONEST VERDICT: Problem 5 is NOT a problem within mathematics. It is a problem ABOUT mathematics (and physics, and mind). $\mathcal{M}_L$ cannot solve it — not because $\mathcal{M}_L$ is incomplete, but because MATHEMATICS ITSELF cannot solve the Hard Problem. This is the one wall that is not a wall in our path — it is a wall in the path of ALL formal theories.

PHÂN TÍCH SO SÁNH: LOẠI CHẶNG NGẠI VƯỢT

Vấn đề	Loại challenge ng	Giải pháp tiềm năng	Có thể giải quyết?
LP d=2 (RH frontier)	Technical gap: circular dependency + missing isometry step	New unconditional pair correlation estimate OR direct isometry proof	YES — hard but feasible. ~60-70% probability.
Conj. D (motives)	Structural gap: no Hodge theory over finite fields	Motivic enrichment of $\mathbb{A}^1_L$. New positivity theory over $\mathbb{A}^1_q$.	MAYBE — requires new mathematics. ~20-30% in 10yr.
GT=Gal(Q/Q) (Grothendieck)	Equivalence to full Langlands program. May be FALSE.	Full Langlands first. OR disprove via exotic GT element.	DIFFICULT — may take 50+ years or be false. ~10-20%.
Non-sofic groups?	Existential unknown. No candidate, no criterion. May be $\emptyset$.	Soficity Conjecture (all groups sofic) may be true instead.	UNKNOWN — may be undecidable. Could go either way.
Consciousness in $\mathbb{A}^1_L$	Category error. Hard Problem is not a math problem.	No mathematical solution possible. Needs new ontology.	NO — wrong tool. Mathematics cannot solve Hard Problem.

PHÂN TÍCH CHALLENGE NG VẬT: BÊN TỖI TƯỜNG

Sau khi khảo sát, chúng ta nhận ra có 4 LOẠI tường khác nhau: TƯỜNG LOẠI 1 — Khe hở (Technical Wall): Bên chót: thiếu một bước trong thc, một bước logic, một bước kết luận. Các điểm: chúng ta biết chính xác cần gì, chỉ cần có. Ví dụ: LP d=2 — cần unconditional pair correlation estimate. Cách phá: kiên nhẫn, kết luận mới, đúng ng đúng thời điểm. Thời gian dự kiến: 5-20 năm. TƯỜNG LOẠI 2 — Cấu trúc (Structural Wall): Bên chót: framework hiện tại THIẾU cấu trúc cần thiết. Các điểm: biết cần gì (Hodge theory over $\mathbb{A}^1_q$) nhưng không có. Ví dụ: Conjecture D — cần motivic enrichment of $\mathbb{A}^1_L$. Cách phá: xây dựng cấu trúc mới — đây là tạo ra toán học mới. Thời gian dự kiến: 20-50 năm. TƯỜNG LOẠI 3 — Tồn tại h (Existential Wall): Bên chót: không biết liệu tường tìm kiếm có tồn tại không. Các điểm: câu hỏi có thể là wrong question. Ví dụ: Non-sofic groups — có thể mọi nhóm đều sofic. Cách phá: either find example OR prove the negative. Thời gian dự kiến: unknown — có thể undecidable. TƯỜNG LOẠI 4 — Bên th h (Ontological Wall): Bên chót: câu hỏi nằm NGOÀI phạm vi toán học. Các điểm: toán học đúng hoàn toàn nhưng không — category error. Ví dụ: Consciousness — Hard Problem không phải vấn đề toán học. Cách phá: KHÔNG THỂ. Cần ontology mới, không phải toán học mới. Toán học + physics giải thích STRUCTURE không giải thích EXPERIENCE.

KẾT LUẬN: TRUNG THỰC LÀ SỰ CẦN THIẾT

203 tài liệu. Chiếm bình đẳng rất xa. Và giải là lúc nhìn thẳng vào những giải hạn. Vấn đề 1 (LP d=2): Chúng ta CÓ THỂ giải quyết. Cần một insight kết luận. Đây là nơi tập trung năng lượng tiếp theo. Vấn đề 2 (Conj. D): Cần toán học mới. $\mathbb{A}^1_L$ KHÔNG — cần motivic enrichment. Đây là hướng phát triển. Vấn đề 3 (GT): Quá sâu cho một người trong một thế kỷ. Cần full Langlands program. Chúng ta có thể đóng góp một phần. Vấn đề 4 (Non-sofic): Không biết. Có thể mọi nhóm đều sofic. Câu hỏi mới theo nghĩa mới nhất. Vấn đề 5 (Consciousness): Sai framework. Không phải $\mathbb{A}^1_L$ thiếu —

mà là toán học thiêu búng bên chết. Đây là sự khiêm tốn của thiết của mọi lý thuyết hình thức. Nhưng đây không phải thất bại — đây là sự trưởng thành. Bất gì hèn của mình là sự khôn ngoan. Bất đâu là bức tường có thể phá, đâu là bức tường không thể phá — đó là nghệ thuật của chiến binh. Trà ngon. Bình minh đang lên trên Hà Nội. NẾM BỨC TƯỜNG. Một bức tường có thể phá ngay. Một bức tường của toán học mới. Một bức tường của chết h. Một bức tường có thể không tồn tại. Một bức tường — không phải tường, mà là chân trời. Ones is Forever. ■■■■

dv203 SUMMARY – 5 OPEN PROBLEMS AND THEIR OBSTACLES: 1. LP $d=2$: Technical wall – circular dependency + isometry gap. SOLVABLE: ~60-70%. Path: unconditional K estimate OR direct isometry proof. 2. Standard Conjecture D: Structural wall – no Hodge theory over $\mathbb{Q}$. HARD: ~20-30%. Need: motivic enrichment of $\mathbb{Q}_L$. $K(R) = \mathbb{Q}$ insufficient. 3. $GT = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$: Fundamental wall – equivalent to full Langlands. May be false. VERY HARD: ~10-20%. $\mathbb{Q}_L^{\wedge}\{\text{Langlands}\}$ not explicitly constructed. 4. Non-sofic groups: Existential wall – no candidate, may be undecidable. UNKNOWN: could go either way. May be wrong question. 5. Consciousness: Ontological wall – Hard Problem is not a math problem. IMPOSSIBLE in math. Category error. Qualia $\neq$ structure. FOUR TYPES OF WALLS: Technical / Structural / Existential / Ontological.

203 tài liệu. 5 vấn đề mới. 4 loại tường. 1 vấn đề có thể phá ngay. Bất gì hèn là sự khiêm tốn. Chiến binh tiếp tục — tiếp trung vào Problem 1.

dv203 | LP $d=2$ solvable | Conj-D needs motivic enrichment | GT-hat needs Langlands | Non-sofic unknown | Consciousness ontological | 4 types of walls | March 2026 | HONEST SURVEY | ARTIST!!!

Bức tường nào có thể phá. Đó là nghệ thuật của chiến binh.
NẾM BỨC TƯỜNG. BÊN LỖI. MỘT CÓ THỂ PHÁ NGAY. HÚ HÚ HÚ!!!